Representation, Not Metaheuristic, Governs Feasibility-Preserving Multi-Objective Optimization: A Study on United States Employment-Based Visa Allocation

Anonymous Author(s)

Affiliation withheld for double-blind review

Abstract. Many hard constrained combinatorial problems are solved by pairing a metaheuristic with a feasibility-preserving decoder, yet the true driver of performance is seldom isolated. Across *four* structurally distinct multi-objective problems, performance is governed far more by the match between representation and search operators than by the metaheuristic family. We develop this on a real case: allocating roughly 140,000 U.S. employment-based visas per year under a per-country ceiling and a spillover cascade, cast as a three-objective integer program (waiting load, inter-country disparity, and visa waste) under six hard constraints and solved with multi-objective Harris Hawks Optimization (MOHHO). Its core is a smallest-position-value random-key mapping composed with a greedy assignment that satisfies every constraint by construction—no penalties, no repair. A controlled six-method ladder isolates the cause: the decoder alone (random restart) already beats a real-coded NSGA-II, and matching operators to the representation lifts three distinct paradigms (genetic algorithm, decomposition, and swarm) above every random-key method—our representation-matched Discrete-MOHHO being the most stable. We stress-test this against a search-space-collapse confound—non-saturating decoders, reference-point sweeps, and an exact bi-objective MILP front—and find the representation effect robust though amplified by the decoder's budget saturation. Replicated on a knapsack, a traveling salesman, and a flow-shop, a matched operator is the best optimizer on *every* structure, while the collapse of a tuned GA below random sampling proves specific to selection landscapes. Parameters are fixed by a Taguchi design; the recovered front Pareto-dominates the incumbent first-in, first-out policy chiefly by eliminating visa waste, yielding an auditable menu of trade-offs as decision support, not a single recommendation.

Keywords: Multi-objective optimization · Harris Hawks Optimization · Constraint handling · Random-key decoder · Permutation encoding · Taguchi design of experiments · Resource allocation

1 Introduction

Each fiscal year the United States Congress authorizes approximately 140,000 employment-based (EB) immigrant visas, distributed across five categories (EB-1 to EB-5). Two structural rules govern the distribution: a *per-country ceiling* ($P_c \approx$ 25,620 visas, the statutory 7 % cross-preference limit) that no single country may exceed, and a *spillover* mechanism that cascades unused visas across categories (EB-4/5 → EB-1 → EB-2 → EB-3). The incumbent system processes petitions strictly first-in, first-out (FIFO) by priority date. The interaction of these rules makes the incumbent policy dysfunctional: two countries concentrate roughly three-quarters of demand and accumulate waits of 10–13 years, while the clash between per-country and per-category ceilings leaves thousands of visas unused.

This matters beyond accounting. Behind every backlogged petition is a household whose relocation, employment, and family reunification hinge on a priority date, so the allocation rule has direct welfare consequences. The problem admits three legitimate, mutually conflicting goals: serve the longest-waiting petitions (*efficiency*), distribute opportunity evenly across nationalities (*equity*), and waste no visas (*utilization*). No single allocation optimizes all three; the policy-relevant object is the set of best compromises—a Pareto front.

Computing that front is hard for three reasons. First, the allocation is *combinatorial*: it assigns integer visa counts to 105 country×category groups under six simultaneous constraints, and its feasible set generalizes the multidimensional knapsack problem, which is NP-hard [13]. Second, the objectives are genuinely *conflicting*, so the solution is a front rather than a point. Third—and most often underestimated—the *hard constraints* are unforgiving: a continuous population metaheuristic naturally proposes infeasible candidates, and the usual remedies (penalty functions, aggressive repair) either require delicate tuning or distort the very search landscape the metaheuristic explores.

We address these challenges with a multi-objective adaptation of Harris Hawks Optimization (MOHHO) whose centerpiece is a *feasibility-preserving decoder*. Our central contribution is methodological and general: a controlled, mechanistically explained demonstration—across four structurally distinct multi-objective problems—that for decoder-based search the *representation–operator match*, not the metaheuristic family, governs performance; the U.S. visa problem is the lead case study through which we establish and stress-test it. Concretely, this paper makes five contributions: *(i)* a decoder that couples a smallest-position-value (SPV) random-key mapping with a deterministic greedy assignment, guaranteeing all six policy constraints *by construction*—without penalties or repair, and therefore without perturbing the search landscape; *(ii)* a multi-objective HHO that pairs this decoder with a crowding-pruned external archive and applies it to a real, constraint-heavy allocation problem rather than to synthetic benchmarks; *(iii)* a Taguchi L9 orthogonal-array tuning of the population size and iteration budget, replacing empirical parameter choices with a systematic experimental design validated by a confirmation run; *(iv)* a controlled six-method ladder that, by lifting three distinct paradigms (a genetic algorithm (GA), decomposition, and a representation-matched *Discrete-MOHHO* we introduce) to within about 1 % of one another once operators match the representation, shows that the feasibility-preserving decoder and the representation–operator match, not the choice of metaheuristic, govern performance—with a direct operator-order-preservation measurement that explains *why* (interpolating GA operators are near-identity on the decoded permutation); and *(v)* a rigorous empirical protocol (30 runs, omnibus Friedman/Nemenyi tests with effect sizes, domination of FIFO, robustness across five perturbed-demand instances, and a four-structure generalization—knapsack, traveling salesman (TSP), and flow-shop—that delineates which findings transfer, the representation–operator law, and which are landscape-specific, the GA-below-random collapse) that makes every claim reproducible.

2 Background and Related Work

Harris Hawks Optimization. Harris Hawks Optimization (HHO) [14] is a population-based metaheuristic inspired by the cooperative hunting of Harris's hawks. Its defining feature is a continuous transition between exploration and exploitation governed by a single physical quantity, the prey's *escape energy* E, which decays with the iteration count; when $|E| \geq 1$ the hawks explore, and when $|E| < 1$ they exploit through soft or hard besieges, optionally with rapid Lévy-flight dives. HHO was introduced for single-objective continuous optimization and has since been extended to the multi-objective setting.

Multi-objective HHO. Kuşoğlu and Yüzgeç [16] adapt HHO to multiple objectives with an external archive and crowding-based leader selection, and Çerçi and Dönmez [4] combine elite non-dominated sorting with a grid index. More recent work folds elitist non-dominated sorting into HHO directly [15] and enhances its operators for both single- and multi-objective settings [6]. These variants are validated mostly on continuous benchmark functions; their application to a *constrained combinatorial* allocation problem, where feasibility—not just convergence—is the binding difficulty, is comparatively unexplored. Our work occupies that gap.

Bridging continuous search and combinatorial structure. The random-key representation of Bean [2] encodes a permutation as a real vector and recovers it by sorting—the SPV rule. Random keys let a continuous operator move freely in $\mathbb{R}^d$ while a decoder interprets each point as a feasible discrete solution. We exploit exactly this separation of concerns: the hawks search a smooth box-constrained space, and a decoder absorbs all combinatorial structure. The idea has matured into biased random-key genetic algorithms [17] and, most recently, a general random-key optimizer that pairs a problem-specific decoder with *any* metaheuristic [5]—the paradigm we instantiate here for HHO, and which our ladder probes directly by varying the metaheuristic while holding the decoder fixed.

Constraint handling in metaheuristics. Three families dominate (see Rahimi et al. [19] for a recent survey). *Penalty methods* fold constraint violations into the objective, but require problem-specific weights and blur the boundary between feasible and infeasible regions. *Repair operators* project infeasible candidates back into the feasible set, but aggressive repair can collapse diversity and overwrite the stochasticity that drives exploration. *Decoder (feasibility-preserving) approaches* guarantee feasibility by construction, leaving the search landscape intact. We adopt the third family: our greedy decoder makes every decoded candidate satisfy all six constraints exactly, so no penalty weight is ever tuned and no repair is ever applied.

Baseline and methodology. NSGA-II (Non-dominated Sorting Genetic Algorithm II) [9] is the canonical multi-objective evolutionary algorithm and our benchmark; such algorithms are the standard tooling for hard allocation and scheduling problems [24]. For parameter selection we use the Taguchi orthogonal-array design [18], a standard tool for tuning stochastic algorithms with few experiments and increasingly applied to metaheuristics [1]. We assess fronts with the hypervolume indicator [25] and with inverted generational distance and spacing [7], and we test significance with nonparametric procedures [11] and the A_{12} effect size [22].

3 Problem Formulation

Decision variables. The problem is discretized into $G = C \times |\mathcal{J}| = 21 \times 5 = 105$ *groups*, where each group g is a (country c, category j) pair. The decision vector is $\mathbf{x} = (x_1, \ldots, x_{105}),\ x_g \in \mathbb{Z}_0^+$, where x_g is the number of visas assigned to group g. Each group has a demand (backlog) n_g and a *waiting weight* $w_g = t_{\text{now}} - d_g$, the years elapsed since its priority date d_g.

Problem instance. The demands n_g and priority dates d_g form a representative instance calibrated to the *documented aggregate structure* of the U.S. employment-based queue: a total backlog of about 1,800,000 pending cases, with India ($\approx 63\,\%$) and China ($\approx 14\,\%$) together accounting for roughly three-quarters of demand, the remainder a long tail over the other countries, and decade-scale waits concentrated in the oversubscribed India and China EB-2/EB-3 categories [3,21]. We are explicit about provenance: the *country-level totals* are anchored to these published aggregates, but the per-(country, category) split and the priority dates d_g are a plausible *synthetic disaggregation* consistent with them, not an observed per-applicant census—only four high-demand blocs (India, China, the Philippines, and Mexico) have category-level demand grounded directly in published data, and the remainder is estimated. Section 6.5 confirms the method comparison is stable under $\pm 20\,\%$ demand perturbation; we note this probes magnitude error, not structural mis-specification of the disaggregation, so the internal method comparison (identical data for all methods) is what we rely on, and the policy reading is illustrative.

Objectives. We minimize three conflicting objectives:

$$f_1(\mathbf{x}) = \frac{\sum_g (n_g - x_g)\, w_g}{\sum_g n_g} \qquad \text{unserved waiting load (efficiency),} \tag{1}$$

$$f_2(\mathbf{x}) = \max_{c_1, c_2 \in \mathcal{C}} \left|\bar{W}_{c_1} - \bar{W}_{c_2}\right| \qquad \text{maximum inter-country disparity (equity),} \tag{2}$$

$$f_3(\mathbf{x}) = V - \textstyle\sum_g x_g \qquad \text{wasted (unassigned) visas (utilization),} \tag{3}$$

where V is the total available visas and $\bar{W}_c = \big(\sum_{g:\, c(g)=c} x_g w_g\big) / \big(\sum_{g:\, c(g)=c} x_g\big)$ is the visa-weighted mean wait of country c. Thus f_2 measures the gap between the longest- and shortest-waiting nationalities: the smaller f_2, the more equitable the distribution. By convention, a country that receives no visas takes its worst backlog wait as $\bar{W}_c$, so starving a nationality maximally penalizes equity—an unserved country is treated as maximally disadvantaged rather than silently dropped from the disparity. This makes f_2 an implicit anti-starvation pressure; FIFO and all reported equity solutions in fact serve every country, so the convention does not drive any headline result.

Constraints. The feasible set $\mathcal{F}$ is defined by six hard constraints:

R1:	$\sum_g x_g \le V = 140{,}000$	(total visas)
R2:	$\sum_{g:\, c(g)=c} x_g \le P_c = 25{,}620 \ \ \forall c$	(7 % per-country ceiling)
R3:	$\sum_{g:\, j(g)=j} x_g \le K_j^{\mathrm{eff}} \ \ \forall j$	(effective per-category cap)
R4:	$0 \le x_g \le n_g \ \ \forall g$	(do not exceed demand)
R5:	$x_g \in \mathbb{Z}_0^+ \ \ \forall g$	(integrality)
R6:	FIFO order within each (c, j) queue	(seniority within a group)

where K_j^{eff} is the per-category cap after the spillover cascade. The cascade is part of the formulation for generality, but we note it is *inert on the instances studied here*: under the documented backlog every category is oversubscribed (demand exceeds its base cap by 3–22×), so the redistributable slack is zero and $K_j^{\mathrm{eff}} = K_j^{\mathrm{base}}$; spillover would bind only in an undersubscribed regime. The result is a three-objective integer program with six constraints whose feasible space grows combinatorially with G, motivating a metaheuristic rather than exact enumeration.

Pareto optimality. A solution $\mathbf{x}$ *dominates* $\mathbf{x}'$ ($\mathbf{x} \prec \mathbf{x}'$) iff $f_i(\mathbf{x}) \le f_i(\mathbf{x}')$ for all i and $f_i(\mathbf{x}) < f_i(\mathbf{x}')$ for some i. The *Pareto front* is $\mathcal{P}^* = \{\mathbf{x} \in \mathcal{F} \mid \nexists \mathbf{x}' \in \mathcal{F} : \mathbf{x}' \prec \mathbf{x}\}$. MOHHO approximates $\mathcal{P}^*$ through an external archive.

4 Methods

4.1 Multi-objective Harris Hawks Optimization

Each hawk is a position $\mathbf{H} \in \mathbb{R}^{105}$ in the box $[0, 1]^{105}$. At iteration t the escape energy is

$$E = 2E_0\big(1 - \tfrac{t}{T}\big), \qquad E_0 \sim U(-1, 1), \tag{4}$$

Table 1. The six MOHHO movement operators, selected by the escape energy $|E|$ and uniform draws q, r. The archive's leader supplies the prey position.

Op.	Condition	Behavior
OP1	$|E| \geq 1,\ q \geq 0.5$	Exploration from a random hawk in the swarm.
OP2	$|E| \geq 1,\ q < 0.5$	Exploration relative to the population centroid.
OP3	$0.5 \leq |E| < 1,\ r \geq 0.5$	Soft besiege: gradual approach to the prey.
OP4	$|E| < 0.5,\ r \geq 0.5$	Hard besiege: aggressive contraction.
OP5	$0.5 \leq |E| < 1,\ r < 0.5$	Soft besiege with rapid Lévy dives.
OP6	$|E| < 0.5,\ r < 0.5$	Hard besiege with rapid Lévy dives.

which drives the choice among six movement operators selected by $|E|$ and uniform draws $q, r \in U(0, 1)$ (Table 1). The multi-objective adaptation introduces three elements. First, the *leader* ("rabbit") is not a single global best but a solution drawn from the external archive with probability proportional to its crowding distance, which biases search toward sparsely populated regions of the front. Second, an *external archive* stores non-dominated solutions and, when it overflows, is pruned by removing the solution with the smallest crowding distance. Third, the archive is updated after every move using the non-dominance relation. The archive size is treated as a tunable parameter (Section 5).

The Lévy operators OP5/OP6 issue a rapid dive; the canonical HHO evaluates two trial points there and keeps the better. To make the comparison *function-evaluation fair*, we accept a single trial point per hawk per iteration, so MOHHO consumes exactly $N \times T$ evaluations—identical to NSGA-II, random restart, and the permutation-native methods, which is the common budget under which every result in Section 6 is reported.

4.2 A feasibility-preserving decoder

The HHO operators act in the continuous space $\mathbb{R}^{105}$, but the problem is combinatorial with six hard constraints. We bridge the two with the SPV rule [2] followed by a deterministic greedy assignment (Algorithm 1). The SPV rule turns a hawk $\mathbf{H}$ into a permutation $\boldsymbol{\pi}$ by sorting the group indices in ascending order of their key h_g. The greedy decoder then walks the groups in that order and assigns

$$x_g = \min\bigl(n_g,\ V_{\text{rem}},\ P^{\text{rem}}_{c(g)},\ K^{\text{rem}}_{j(g)}\bigr), \tag{5}$$

decrementing the residual budgets of R1–R3 after each step. Because every assignment is capped by the remaining budgets and by demand, and counts are integers, *every* decoded vector satisfies R1–R6 by construction. No penalty term is introduced and no repair is performed, so the hawks search directly on a feasible objective landscape, preserving both the structure of HHO's cooperative foraging and its stochasticity. One property of the greedy fill should be stated plainly: it assigns each visited group its *maximal* feasible amount, so the decoder's image is the set of budget-*saturating* feasible allocations—a strict subset of $\mathcal{F}$,

Algorithm 1 Feasibility-preserving SPV + greedy decoder

Require: hawk $\mathbf{H} \in \mathbb{R}^{105}$; demands n_g; budgets V, $\{P_c\}$, $\{K_j^{\text{eff}}\}$
1: $\boldsymbol{\pi} \leftarrow$ ArgSortAscending($\mathbf{H}$) ▷ SPV: keys → permutation
2: $V_{\text{rem}} \leftarrow V$; $P_c^{\text{rem}} \leftarrow P_c \ \forall c$; $K_j^{\text{rem}} \leftarrow K_j^{\text{eff}} \ \forall j$
3: **for** $g \in \boldsymbol{\pi}$ **do** ▷ visit groups in SPV order
4: $\quad x_g \leftarrow \min\bigl(n_g, V_{\text{rem}}, P_{c(g)}^{\text{rem}}, K_{j(g)}^{\text{rem}}\bigr)$
5: $\quad V_{\text{rem}} \mathrel{-}= x_g$; $P_{c(g)}^{\text{rem}} \mathrel{-}= x_g$; $K_{j(g)}^{\text{rem}} \mathrel{-}= x_g$
6: **end for**
7: **return x** ▷ satisfies R1–R6 without repair

not all of it. Every dominance and Pareto statement below is therefore relative to this decoder-reachable set. This is the right object here rather than a limitation: because visa waste f_3 is itself a minimized objective, deliberately reserving budget while applicants wait is rarely attractive; an exact bi-objective (f_1, f_3) MILP (Section 6.4) confirms that the few non-saturating non-dominated allocations that do exist improve no objective by a practically meaningful margin, so restricting to this subset loses nothing relevant.

Proposition 1 (Feasibility by construction). *For any hawk* $\mathbf{H} \in \mathbb{R}^{105}$*, the allocation* $\mathbf{x}$ *returned by Algorithm 1 satisfies all six constraints R1–R6.*

Proof. The residuals are initialized to V, P_c, $K_j^{\text{eff}} \geq 0$ and, at each step, decremented by $x_g = \min(n_g, V_{\text{rem}}, P_{c(g)}^{\text{rem}}, K_{j(g)}^{\text{rem}})$. By induction each residual stays ≥ 0, since x_g never exceeds the current residual; hence $0 \leq x_g \leq n_g$ (R4) and, as the minimum of nonnegative integers, $x_g \in \mathbb{Z}_0^+$ (R5). Because $x_g \leq V_{\text{rem}}$ and V_{rem} is the unconsumed total, $\sum_g x_g \leq V$ (R1); likewise $x_g \leq P_{c(g)}^{\text{rem}}$ and $x_g \leq K_{j(g)}^{\text{rem}}$ keep the per-country and per-category sums within P_c (R2) and K_j^{eff} (R3). Finally, each group is a single (c, j) pair sharing one priority date, so assigning the count x_g serves its x_g most-senior petitions, honoring FIFO order within the queue (R6). Thus $\mathbf{x} \in \mathcal{F}$ for every input, with no penalty term and no repair. □

4.3 Benchmarks and a representation-matched optimizer

To separate the contribution of the decoder, the representation, and the search heuristic, we compare a *ladder* of methods that all share the same greedy decoder, objectives, hypervolume reference point, and 50 × 500 budget; they differ only in how they explore. Two operate on the continuous SPV random keys in $\mathbb{R}^{105}$: *(a)* the MOHHO of Section 4.1; and *(b)* **NSGA-II** [9] with simulated binary crossover (SBX, $\eta_c = 20$, $p_c = 0.9$), polynomial mutation ($\eta_m = 20$, $p_m = 1/d$), binary tournament selection, and elitist $(\mu+\lambda)$ replacement. A third, *(c)* **random restart**, draws the 25,000 candidates uniformly at random (no search), isolating what the decoder alone achieves.

The ablation (Section 6.4) reveals that the continuous random-key operators are a poor match for the induced permutation landscape. We therefore also evaluate three **permutation-native** methods—spanning three distinct paradigms—that act directly on permutations of the 105 groups (decoded by the same greedy procedure): *(d)* **permutation-NSGA-II** (dominance-based) with order crossover (OX) and swap mutation; *(e)* **permutation-MOEA/D** [23] (decomposition-based), with Tchebycheff scalarization over a simplex-lattice of weight vectors and the same OX + swap operators; and *(f)* **Discrete-MOHHO**, our representation-matched Harris Hawks optimizer. Discrete-MOHHO keeps HHO's signature—the escape energy E of Eq. (4) scheduling exploration and exploitation, and a crowding-pruned external archive supplying the leader—but expresses every move on permutations: when $|E| \geq 1$ a hawk explores by recombining (OX) with a random hawk or by a heavy random shuffle; when $|E| < 1$ it besieges the leader by inheriting an order segment from it (OX) and perturbing with a number of swaps that shrinks as $|E| \to 0$, with occasional Lévy-like segment reversals. All six methods run for 30 independent seeds under the identical budget.

4.4 Evaluation protocol and indicators

We assess each approximation front with four indicators: the *hypervolume* (HV; larger is better) dominated with respect to the reference point (10; 16; 20,000), an upper bound on each objective; the *inverted generational distance* (IGD; smaller is better) to a reference front Z formed by the non-dominated union of the MOHHO and NSGA-II combined fronts ($|Z| = 113$; because Z is built from the compared methods' own fronts, we treat IGD as an internal convergence/coverage indicator rather than distance to an independent ground truth, and rest the headline conclusions on the reference-point-robust HV); the *spacing* indicator (smaller is better) measuring uniformity; and the *cardinality* of the non-dominated set. Objectives are min–max normalized before IGD and spacing. Each algorithm is run for 30 independent seeds, and a *combined front* is formed as the non-dominated union of its 30 per-run fronts. Statistical significance between the two 30-value HV distributions is assessed with a Mann–Whitney U test [11], and effect size with the Vargha–Delaney A_{12} [22]; where one-sided p-values are reported the alternative direction is fixed a priori from the mechanism (decoder- and representation-matched methods are predicted to improve over the real-coded baseline), and the two-sided values lead to the same conclusions at these effect sizes. The deterministic **FIFO** baseline orders groups by ascending priority date and is decoded with the same greedy procedure, yielding $f_1 = 8.7891$, $f_2 = 13.0$ years, and $f_3 = 1{,}940$ wasted visas.

5 Parameter Tuning by Taguchi Orthogonal Array

Population-based metaheuristics are sensitive to their control parameters, and choosing them by trial and error is neither systematic nor reproducible. We therefore tune the two parameters that most directly govern the search effort—the population size N and the iteration budget T—together with two secondary parameters, the external-archive size and the Lévy exponent β, using a Taguchi

Table 2. L9(3^4) orthogonal array, configurations, and response (12 replications per row). HV is reported as mean ± s.d. in units of 10^6; S/N is the larger-the-better ratio of Eq. (6).

Run	$\boldsymbol{N}$	$\boldsymbol{T}$	**Archive**	β	**HV** ($\times 10^6$)	**S/N (dB)**
1	30	100	50	1.3	0.290 ± 0.010	109.220
2	30	300	100	1.5	0.296 ± 0.005	109.419
3	30	500	150	1.7	0.292 ± 0.010	109.304
4	50	100	100	1.7	0.293 ± 0.007	109.323
5	50	300	150	1.3	0.297 ± 0.010	109.446
6	50	500	50	1.5	0.301 ± 0.006	109.560
7	70	100	150	1.5	0.301 ± 0.008	109.574
8	70	300	50	1.7	0.303 ± 0.010	109.603
9	70	500	100	1.3	0.306 ± 0.008	109.719

orthogonal-array design [18]. The Taguchi method probes a four-factor space with a small, balanced set of experiments and quantifies each factor's influence through a signal-to-noise (S/N) ratio that rewards both a high mean response and low variance across replications.

Design. We use four factors at three levels each (Table 2). The response is the hypervolume HV of the final archive; because larger HV is better, we use the larger-the-better S/N ratio

$$\mathrm{S/N} = -10 \log_{10}\left(\frac{1}{r}\sum_{k=1}^{r} 1/y_k^2\right), \tag{6}$$

where y_k is the HV of replication k. A full-factorial design would need $3^4 = 81$ configurations; the L9(3^4) orthogonal array estimates all four main effects with just nine. Each configuration is replicated with $r = 12$ independent seeds drawn disjointly from the 30-run benchmark seeds, so the tuning data never contaminate the evaluation of Section 6.

Main effects. Averaging the S/N ratio over the levels of each factor gives the response table (Table 3) and the main-effects plot (Figure 1). The ranking by S/N range is unambiguous: *population size* ($\Delta = 0.318$ dB) and *iteration budget* ($\Delta = 0.155$ dB) dominate, whereas the Lévy exponent ($\Delta = 0.108$ dB) and the archive size ($\Delta = 0.046$ dB) lie within the run-to-run noise. This is itself an informative result—it confirms that the two parameters flagged as needing justification are precisely the influential ones, and that the algorithm is robust to the secondary choices. The population effect is monotone and grows toward the top of the range (largest at $N = 50 \to 70$, +0.189 dB), whereas the iteration effect saturates (+0.117 dB from T=100 to 300, then only +0.039 dB to 500), confirming that the iteration budget is past its point of diminishing returns.

Optimum and confirmation. Selecting the best level of each factor yields the predicted optimum $N = 70$, $T = 500$, archive = 100, $\beta = 1.5$. Under the

Table 3. Response table: mean S/N (dB) per factor level, range Δ, and rank. Larger Δ means a more influential factor; the highest-S/N level in each row is in bold. Each Δ is computed from unrounded S/N values.

Factor	Level 1	Level 2	Level 3	Δ	Rank
A: population N	109.314	109.443	**109.632**	0.318	1
B: iteration budget T	109.372	109.489	**109.528**	0.155	2
C: archive size	109.461	**109.487**	109.441	0.046	4
D: Lévy exponent β	109.462	**109.518**	109.410	0.108	3

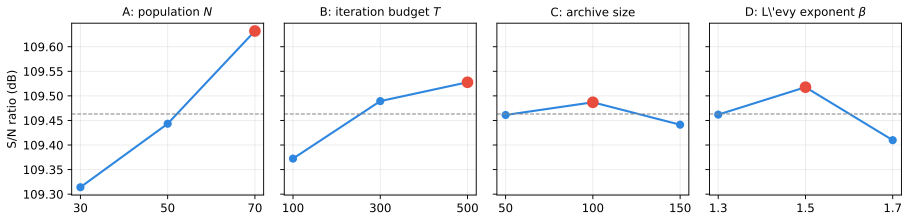
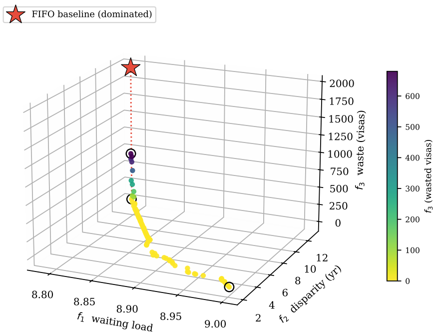


Fig. 1. Main-effects plot of the S/N ratio. The dashed line is the grand mean (109.46 dB) and the red marker the best level per factor. Population and iteration budget show clear trends (population the steeper); archive size and β are essentially flat.

additive model, its predicted S/N is $\approx$ 109.78 dB. A *confirmation run* at that configuration (12 fresh seeds) gives an observed S/N of 109.66 dB (HV = 304,391). The $\approx$ 0.12 dB gap between prediction and observation is small—the slight positive bias is the expected signature of an additive model that ignores factor interactions—so the additive assumption holds and the tuning is internally consistent.

Adopted operating point. The 30-run study and the NSGA-II benchmark of Section 6 use $N = 50$, $T = 500$, archive = 100, and $\beta = 1.5$. A confirmation run at this configuration gives S/N = 109.56 dB (HV = 300,775), within 1.2 % hypervolume of the L9 optimum. We retain it deliberately: its *sole* difference from the optimum is the population size (50 vs. 70)—T, archive size, and β all match the optimum—and increasing N from 50 to 70 raises the per-iteration cost by 40 % for only a $\approx$1.2 % HV gain. Likewise $T = 500$ is intentionally conservative—it exceeds by a wide margin the empirical 99 %-convergence point at iteration 138 (Section 6.2)—so the budget cannot be blamed for any shortfall. In short, the Taguchi study turns a previously empirical choice into a justified operating point about a decibel-tenth below the L9 optimum, with the dominant factor (T) already at its best level.



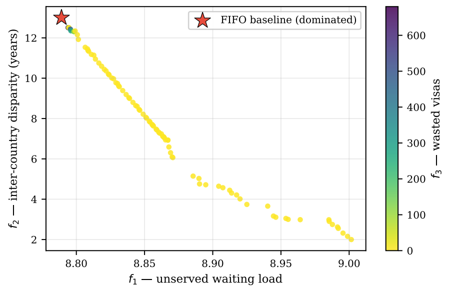


Fig. 2. The 92-solution combined Pareto front over the three objectives, colored by the waste f_3. The FIFO baseline (red star) floats above the front—the dotted drop-line marks its distance to the front's waste level—and is dominated; circles mark the extreme solutions.

Fig. 3. f_1–f_2 projection of the same front, colored by f_3 (wasted visas). FIFO lies inside the dominated region.

Table 4. Extreme solutions of the combined front, compared with the FIFO baseline. The minimum-f_1 solution dominates FIFO (better on two objectives, equal on the already-maximal disparity).

Solution	f_1	f_2 **(yr)**	f_3 **(visas)**	**Remark**
Min. f_1 (least waiting)	8.7884	13.0	680	**Dominates FIFO**
Min. f_2 (most equitable)	9.0016	2.0	0	Radical equity
Min. f_3 (full use)	8.851	8.06	0	100 % utilization
FIFO baseline	8.7891	13.0	1,940	Reference (dominated)

6 Results

6.1 Combined Pareto front and domination of FIFO

Combining and re-filtering the 30 runs yields a front of **92 non-dominated solutions**. Figure 2 shows it in the full objective space and Figure 3 in the f_1–f_2 projection; in both, the FIFO baseline floats *above* the front, visual evidence that it is dominated. The extreme solutions (Table 4) expose the trade-offs.

The central empirical finding is that **FIFO is not Pareto-optimal**: the minimum-f_1 solution Pareto-dominates it—strictly better on waste (f_3 from 1,940 to 680 visas), tied on the already-maximal disparity ($f_2 = 13.0$), and negligibly better on the near-degenerate f_1 (8.7891 to 8.7884; the f_1 range is so narrow—Section 6.2—that this edge carries no practical weight). The substantive case against FIFO is therefore about *waste and equity*, not waiting load: 82 of the 92 front solutions reach $f_3 = 0$ (zero waste, which FIFO never attains), and the front spans disparities all the way from FIFO's 13.0 years down to 2.0. (The

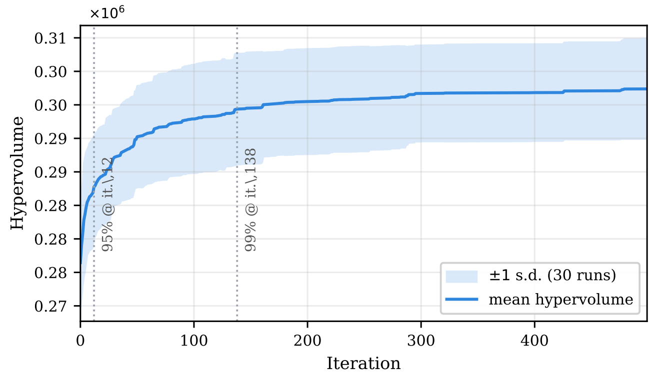


Fig. 4. Mean hypervolume over 30 runs with a ±1 standard-deviation band. The dotted markers indicate the iterations at which 95 % and 99 % of the final HV are attained; gains beyond iteration 140 are marginal.

MOHHO, Discrete-MOHHO, and permutation-NSGA-II combined fronts agree within 0.2 % in hypervolume (Table 7), so the policy menu does not hinge on the optimizer choice—the recommended single-run optimizer, Discrete-MOHHO, yields an essentially equivalent front.)

6.2 Convergence and robustness

The mean hypervolume curve (Figure 4) reaches 95 % of its final value by iteration 12 and 99 % by iteration 138; gains beyond iteration 140 are under 1 %. Over the 30 runs the HV distribution is narrow (mean 302,379, standard deviation 7,455), a coefficient of variation (CV) of **2.47 %**, evidence of a stable and reproducible algorithm.

6.3 Comparison with NSGA-II

We first compare the two *random-key* optimizers under the identical problem, encoding, decoder, and budget of Section 5. MOHHO outperforms the standard (real-coded) NSGA-II on the aggregate front-quality measures (Table 5)—the two tie on the extreme values the decoder makes attainable for both (best f_1 and zero waste). MOHHO attains a higher mean hypervolume (+3.1 %), a smaller IGD, and more uniform spacing; the gap is significant (one-sided Mann–Whitney $p \approx 1.3 \times 10^{-5}$, $A_{12} = 0.82$). Figure 5 shows the front coverage. This advantage is a property of the *search dynamics*, not of archiving (granting NSGA-II the identical archive barely moves it; Section 6.4): MOHHO's position updates induce genuine permutation jumps on the random keys, whereas NSGA-II's interpolating operators scarcely perturb the decoded order (Section 6.4 quantifies this). That comparison, however, is only part of the story.

Table 5. MOHHO vs. NSGA-II under an identical problem, encoding, decoder, and budget (30 runs each). Best value per row in bold. The FIFO point is dominated by both and is not a front.

Metric	FIFO	MOHHO	NSGA-II
Method type	deterministic	metaheuristic	metaheuristic
Best f_1 (waiting load)	8.7891	8.7884	8.7884
Best f_2 (disparity, yr)	13.0	**2.0**	2.25
Best f_3 (waste, visas)	1,940	0	0
Mean HV $\pm\sigma$ (30 runs)	—	**302,379 ± 7,455**	293,367 ± 6,996
Coefficient of variation	—	2.47 %	**2.38 %**
Combined-front HV	—	**321,446**	316,060
Non-dominated solutions	1	92	92
IGD vs. Z (smaller better)	—	**0.0052**	0.0080
Spacing (smaller better)	—	**0.0099**	0.0463
Significance vs. MOHHO	dominated	—	$p \approx 1.3 \times 10^{-5}$

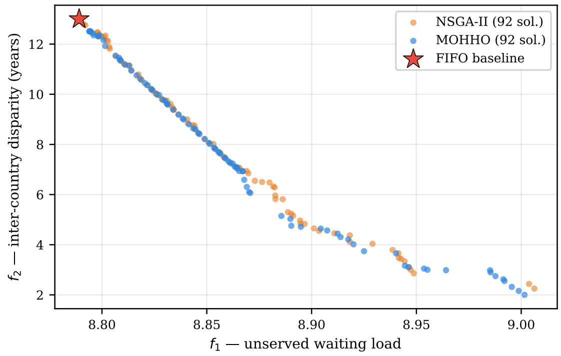


Fig. 5. Combined fronts of MOHHO and (real-coded) NSGA-II in the f_1–f_2 projection. MOHHO covers the front more densely and more uniformly.

6.4 Decoder, representation, and search: a controlled ladder

Section 6.3 compared two random-key methods. To attribute performance to its true source we now climb a ladder of methods, all sharing the decoder and a 25,000 function-evaluation budget—the comparison currency (Table 7). The permutation-native methods inherit the Taguchi-tuned N and T and are not separately tuned, so any edge they show is not a tuning artifact.

The decoder does most of the work. *Random restart*—blind sampling with no search—already reaches a mean HV of 309,821: 5.6 % *above* NSGA-II and, strikingly, 2.5 % *above* classic MOHHO itself (the swarm wins on only 9/30 paired seeds). Under an identical function-evaluation budget the feasibility-preserving decoder shapes a landscape on which blind sampling outperforms *both* real-coded searches; giving NSGA-II the *identical* archive barely helps it (293,666 vs.

293,367), so NSGA-II's deficit is not an archiving artifact. The swarm's loss is likewise not an artifact of our single-trial-point budgeting (Section 4.1): a control with the *canonical* two-point Lévy dive does not beat random restart either—at the matched 25,050-evaluation budget it reaches 308,926 ($-0.3\,\%$ vs. random restart, paired Wilcoxon $p = 0.77$), and even granted its native schedule ($\approx$35,600 evaluations, 42 % over budget) it only *ties* random restart ($+0.4\,\%$, $p = 0.21$). The swarm does not beat blind sampling under either dive convention.

The representation–operator match is decisive. Why does a tuned NSGA-II fall below blind sampling? We measured the cause directly. On the SPV random keys, SBX and polynomial mutation are *near-identity on the decoded permutation*: over 3,000 trials the Kendall rank correlation between a parent's SPV order and its child's is $\tau = 0.99$. This is not an artifact of measuring on random parents or of one operator setting: measured *on-trajectory*—parents drawn by the algorithm's own tournament from live populations at iterations 1 through 499—the correlation stays in $[0.987, 0.992]$ throughout the run; and sweeping the crossover distribution index $\eta_c \in \{2, \ldots, 100\}$ (Figure 6) shows that even the most disruptive crossover (η_c=2, τ=0.92) leaves the real-coded GA's mean HV at 305,293—still below blind random restart (309,821) and far below the permutation tier—while across the sweep order-preservation and HV are perfectly anti-correlated (Kendall -1.0, $p = 0.003$). No crossover spread lets a random-key GA escape the random-key ceiling; only matching the operators to the representation does. The GA perturbs the keys so little that the induced permutation barely changes, leaving it almost frozen in combinatorial space—precisely why it sinks below blind sampling, which at least draws fresh permutations. HHO's position updates, by contrast, *propose* large permutation jumps ($\tau \approx 0$); the dominance-gated acceptance rarely admits them—only about 0.6 % of hawks move per iteration, so the realized population is largely frozen—but the external archive is fed by these rejected proposals, so the archived front still samples the decoded space far more widely than NSGA-II's near-identity proposals (hence MOHHO > NSGA-II within the random-key tier). The real-coded MOHHO is thus a deliberately standard, *weak* baseline: on a concave benchmark front (ZDT2) its operators collapse the whole population to a corner—hypervolume only 0.11 of the true front under every acceptance rule and every reasonable repair we tried—so our thesis rests not on the swarm being strong but on the opposite, that real-coded search loses to permutation-native operators. The clean fix, though, is to match the operators to the representation. Acting *directly* on permutations lifts *three different paradigms* above every random-key method: permutation-NSGA-II (dominance/GA) reaches 318,151, permutation-MOEA/D (decomposition) 314,846, and our **Discrete-MOHHO** (swarm) 316,637. The three cluster within about 1 % of one another, yet all sit 1.6–2.7 % above the best random-key method (random restart)—more than the spread *among* the matched paradigms. Discrete-MOHHO improves on classic MOHHO by $+4.7\,\%$ (paired Wilcoxon $p = 9.3 \times 10^{-10}$, better on all 30/30 seeds) and is the *most stable* method (CV 0.54 %, versus 0.58 % for permutation-NSGA-II and 2.47 % for classic MOHHO), making it the most

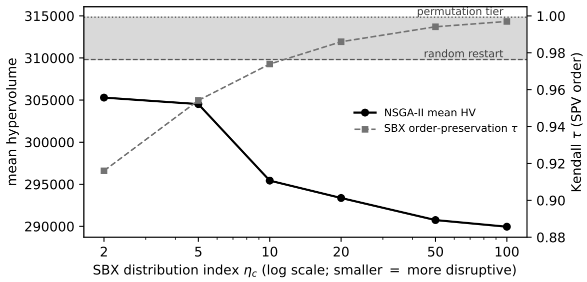


Fig. 6. Sweeping the SBX crossover spread η_c for the real-coded NSGA-II on the visa problem. As η_c falls (more disruptive crossover) the operator order-preservation τ drops, yet the mean hypervolume never reaches blind random restart (dashed) let alone the permutation tier (dotted); across the sweep τ and HV are perfectly anti-correlated (Kendall -1.0). No crossover spread lets a random-key GA escape the random-key ceiling—only matching operators to the representation does.

dependable choice for a single-run decision. The marginal mean lead belongs to permutation-NSGA-II (+0.48 % over Discrete-MOHHO, $A_{12} = 0.27$).

Operator and metaheuristic family are both second-order. The three matched paradigms above share the OX + swap operators, so that comparison varied the *selection/replacement architecture* while holding the operator fixed. To probe the operator axis directly we ran a 3×3 factorial—three operator sets (OX + swap, PMX (partially-mapped crossover) + insertion, CX (cycle crossover) + inversion) crossed with the three architectures, 30 seeds each (Table 6). A two-way variance decomposition attributes just 5.5 % of the hypervolume variance to the operator set and 13.4 % to the architecture (interaction 2.9 %), against 78.3 % from seed-to-seed noise; switching either axis moves the mean by at most ∼2 % (operator marginal range 4,139, architecture range 6,720 HV), an order of magnitude below the run-to-run variation. So *once the representation is matched, neither the operator set nor the metaheuristic family governs performance*: the representation–operator match—the random-key→permutation jump—is first-order, while both design axes *within* the matched representation are second-order. Discrete-MOHHO is the most operator-robust of all, holding ≈317,000 HV across every operator set. Architecture is the larger of the two second-order effects (13.4 % vs. 5.5 %)—and Discrete-MOHHO's energy-besiege architecture is part of why we recommend it—but both remain an order of magnitude below the seed noise, so each is second-order relative to the first-order representation jump.

An empirical regularity. The ladder (Figure 7) makes the pattern unmistakable: performance is governed by the *feasibility-preserving decoder* and the *representation–operator match*, not by the metaheuristic family. That encoding–operator mismatch can hurt search is known in principle since random keys were introduced [2]; our contribution is the *controlled, quantified* demonstra-

Table 6. Operator × architecture factorial on the visa problem: mean hypervolume (30 seeds) for three permutation operator sets crossed with three selection architectures, all under the matched budget. A two-way decomposition gives $\eta^2 = 5.5\,\%$ (operator), 13.4 % (architecture), 2.9 % (interaction), 78.3 % (seed noise): both design axes are second-order once the representation is matched.

Operator set	perm-NSGA-II	perm-MOEA/D	Discrete-MOHHO
OX + swap	317,237	313,608	317,569
PMX + insertion	311,284	309,316	317,715
CX + inversion	309,702	309,248	317,048

tion on a real constrained problem—including the counterintuitive result that *both* real-coded searches (a tuned NSGA-II and the MOHHO swarm) fall below blind sampling—and the decomposition of performance into decoder, representation, and family. That three *distinct* paradigms (swarm, decomposition, dominance/GA) converge to within about 1 % once the representation is matched, all far above the random-key methods, is strong evidence that the family is second-order here. We are precise about what this shows: the operator–representation *match* is first-order—the random-key→permutation jump—whereas *within* a matched representation both the operator set and the selection/replacement architecture (Pareto sorting, scalarized decomposition, or energy-driven besiege) are second-order, the factorial of Table 6 attributing only 5.5 % and 13.4 % of the variance to them. The matched swarm reclaims competitiveness and the best stability, which is why we promote Discrete-MOHHO. An omnibus test across the six methods on a common 30-seed set—a paired Friedman test with a Nemenyi post-hoc [10], identical to the analysis of Section 6.6—rejects equal performance decisively ($\chi^2 = 112.6$, $p \approx 1.2 \times 10^{-22}$, critical difference 1.38), and the mean ranks split into two tiers: the three permutation-native methods (ranks 1.6–2.5) above the three random-key methods (4.2–5.8), with NSGA-II worst and every permutation-native method significantly above random restart. Finer within-tier separations fall within the Nemenyi critical difference, consistent with the "within about 1 %" clustering. The ranking is also robust to the hypervolume reference point: across five reference points random restart outranks both real-coded searches throughout, and the strict permutation/random-key tier split holds under four of the five, breaking only at the tightest. We frame this as an empirical regularity for this problem, not a general law: all instances here share one structural skeleton—which the structurally distinct problems of Section 6.6 directly address.

Robustness to a search-space-collapse confound. A skeptic could argue that the decoder's budget saturation collapses the search space so far that the optimizer is necessarily second-order. We tested this on three axes. *Structure*: 50,000 random permutations decode to 39,401 distinct objective points (degeneration ratio 1.27), and a principal-component analysis (PCA) places about 99 % of the front's variance in two components—the first component alone captures only 0.81–0.91,

Table 7. The method ladder: 30 runs each, identical greedy decoder and 25,000-evaluation budget. Top block: random-key encoding. Bottom block: permutation-native, spanning three distinct paradigms (swarm, decomposition, dominance/GA), which cluster within about 1 % and dominate every random-key method. Best per column in bold; "perm-" abbreviates "permutation-".

Method	Paradigm	Mean HV $\pm \sigma$	CV	Comb. HV	# sols
NSGA-II	GA	$293{,}367 \pm 6{,}996$	2.38 %	316,060	92
MOHHO	swarm	$302{,}379 \pm 7{,}455$	2.47 %	321,446	92
Random restart	—	$309{,}821 \pm 2{,}513$	0.81 %	316,383	81
Discrete-MOHHO	swarm	$316{,}637 \pm 1{,}720$	**0.54 %**	321,354	**149**
perm-MOEA/D	decomp.	$314{,}846 \pm 5{,}384$	1.71 %	320,969	90
perm-NSGA-II	GA	$\mathbf{318{,}151 \pm 1{,}855}$	0.58 %	**321,935**	148

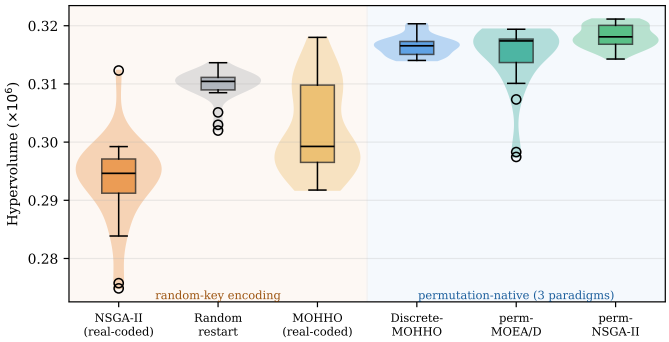


Fig. 7. The method ladder: per-run hypervolume (30 runs each, identical decoder and budget). Crossing from random-key encodings (left, shaded amber) to permutation-native operators (right, shaded blue) produces a clear jump; the metaheuristic family (GA vs. Harris Hawks) matters far less than the representation. Discrete-MOHHO is the tightest distribution (most stable).

so the effective problem is *near-mono-objective*—one wide axis (f_2, $\sim$39$\times$ wider than f_1 across these random decodes) with a genuine but narrow second axis, a deliberately weak multi-objective lead case. *Decoder*: re-running the full ladder on two *non-saturating* decoders (a fractional-intensity and a stochastic-skip decoder, both feasible by construction) keeps the permutation tier significantly above the random-key tier under *all three* decoders (paired Wilcoxon $p < 10^{-3}$) and never inverts it; the margin shrinks from 6.9 % (greedy) to 4.3 % and 1.4 %, so saturation *amplifies*—it does not create—the effect, and an exact (f_1, f_3) mixed-integer linear program (MILP) front (with f_2 a posteriori) exposes no practically meaningful non-saturating solution the greedy front misses. *Metric*: the GA operators' near-identity holds on the full decoded *allocation*, not just the order (an SBX step barely perturbs the decoded vector—consistent with its order near-identity $\tau = 0.99$—whereas an HHO step moves it substantially), and the per-run tier ranking is reference-point-robust (above)—though on the *combined* fronts the strict tier softens under reference-free metrics (IGD$^+$, additive ϵ), which converge to a near-common reference set, so the tier is a per-run phenomenon, and the hypervolume that exhibits it is dominated by the f_2 axis—the only objective with a wide range, f_1 being more than an order of magnitude narrower and f_3 saturating. The confound thus *bounds* rather than overturns the claim: the effect is real and robust, modest on this saturation-amplified, f_2-dominated instance, with the general law resting on the four structures below.

6.5 Generalization across instances

To check that these findings are not specific to a single backlog estimate, we built five further instances by perturbing every group's demand n_g with an independent uniform $\pm 20\,\%$ jitter (fixed seeds) and recomputing the spillover-adjusted category caps and the per-country caps accordingly; on each we re-ran the full comparison (10 seeds per method). The pattern is invariant (Figure 8): on all five perturbed instances—as on the base instance—the combined front dominates FIFO, MOHHO outperforms NSGA-II (one-sided Mann–Whitney $p \leq 0.071$ in every case), and the decoder-dominance signature persists, with random restart matching or exceeding MOHHO (101–103 % of its hypervolume) and both above NSGA-II (which trails at 96–99 % of MOHHO). The pattern therefore holds across six instances, indicating that the conclusions are robust to $\pm 20\,\%$ demand-estimation error rather than artifacts of one estimate. We are careful not to overstate this: all six instances share the same structural skeleton (21 countries, 5 categories, the same caps and the demand-far-exceeds-supply regime), so this is robustness to demand perturbation; generalization across structurally distinct problems is addressed directly in Section 6.6.

6.6 Generalization across structurally distinct problems

The instances above share one structural skeleton. To test whether the regularity reaches genuinely different structures, we replicate the entire six-method ladder on *three* further classic multi-objective combinatorial problems that share nothing with visa allocation but the SPV + greedy/permutation methodology: a 0/1 **multidimensional knapsack** (MOMKP; 120 items, four capacities—a *selection* landscape; its ladder is shown in Figure 9), a tri-objective **traveling salesman** (mo-TSP; 100 cities, three cost matrices—a cyclic *sequencing* landscape), and a tri-objective **permutation flow-shop** (mo-PFSP; 50 jobs, 10 machines; makespan, flowtime, and tardiness—a *scheduling* landscape). Each runs the same six methods, budget, and 30 common seeds, analyzed by a paired Friedman test with a Nemenyi post-hoc (the Demšar standard [10]; Table 8).

One result is *robust across all four structures*: the best optimizer is *always* a permutation-native method (perm-NSGA-II on the visa, knapsack, and flow-shop problems; perm-MOEA/D on the TSP), and neither blind random restart nor any real-coded method is ever best. Matching the operators to the representation reliably produces the winner, and the metaheuristic family stays second-order

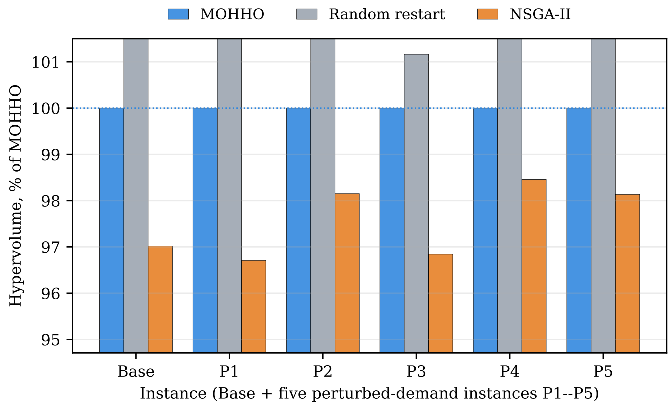


Fig. 8. Generalization over six instances (base plus five ±20 % perturbed-demand instances, P1–P5), each method's mean hypervolume shown as a percentage of MOHHO's on that instance. The ranking MOHHO > random restart > NSGA-II is preserved throughout, and FIFO is dominated on every instance.

among the matched methods. All four omnibus tests reject equality decisively (Friedman χ^2 from 112 to 150, $p < 10^{-21}$, critical difference 1.38). Our **Discrete-MOHHO** is top-two on three of the four structures and the most stable of the matched methods, which is why we recommend it for single-run decisions even where perm-NSGA-II edges it on the mean.

What does *not* generalize is the dramatic version of the story—a tuned real-coded GA collapsing *below* blind random sampling. This holds on the two *selection* landscapes (visa and knapsack, where the real-coded NSGA-II is the worst method, beneath random restart) but reverses on the two *sequencing* landscapes: on the TSP and the flow-shop the real-coded NSGA-II is competitive (mean ranks 3.0 and 2.7) and *random restart is instead the worst* method. The mechanism reconciles this. Operator order-preservation is problem-independent—we re-confirmed τ on each problem's keys (SBX $\approx$0.99, HHO $\approx$0)—but its *consequence* is landscape-dependent: near-identity SBX leaves the decoded order almost frozen, which is fatal on selection landscapes (where blind sampling at least draws fresh subsets) yet acts as effective *local search* on sequencing landscapes (where small reorderings are precisely the useful neighborhood and blind sampling of full permutations is poor). We also asked whether a single "search-headroom" scalar predicts which structures show the collapse; a five-point capacity sweep gives it no support (Spearman $\rho = -0.90$, but $n = 5$: exact two-sided $p = 0.083$, inconclusive), so the exact landscape determinant remains open. The transferable claim is therefore narrow and precise: *representation-matched operators yield the best optimizer on every structure, and the metaheuristic family is second-order among them*; the spectacular "GA-below-random" collapse is a property of selection/assignment landscapes, not a universal law. We report this scoping rather than the punchier over-generalization.

Table 8. Mean Friedman rank (lower is better; 30 common seeds, paired) of each method on four structurally distinct multi-objective problems. A permutation-native method is best on every structure (**bold**). The real-coded NSGA-II collapses below random restart only on the *selection* landscapes (visa, knapsack); on the *sequencing* landscapes (TSP, flow-shop) it is competitive and random restart is the worst method instead. "rk" = random-key encoding, "pm" = permutation-native.

Method	Enc.	Visa	Knapsack	TSP	Flow-shop
perm-NSGA-II	pm	**1.60**	**1.10**	2.00	**1.80**
perm-MOEA/D	pm	2.53	2.87	**1.00**	2.83
Discrete-MOHHO	pm	2.23	2.10	4.00	2.63
MOHHO (real-coded)	rk	4.67	3.93	5.00	5.07
Random restart	rk	4.20	5.23	6.00	5.93
NSGA-II (real-coded)	rk	5.77	5.77	3.00	2.73

7 Discussion

The trade-off is structural. The conflict among objectives is not an artifact of the model. Two countries concentrate roughly three-quarters of demand with 10–13-year waits; the 7 % ceiling forbids fully serving them, so reducing disparity (f_2) forces unserved load to remain in those countries ($f_1 \uparrow$), and distributing equitably leaves idle capacity in saturated categories ($f_3 \uparrow$). The three extreme solutions of Table 4 are mutually non-dominated—empirical proof of genuine three-way conflict.

Sensitivity. Two observations bound the result. First, the natural front holds only about 38 solutions per run—well under the size-100 archive cap, so pruning never activates and the archive size is immaterial (hypervolume unchanged from 100 to 500). Second, the range of f_1 across the front (0.21, vs. f_2's $\approx$11 years) is more than an order of magnitude narrower: because total demand (1,800,000) far exceeds the 140,000 visas, almost any permutation leaves a similar unserved load, so f_1 variability is constrained by the *problem structure*, not by the algorithm. We nonetheless keep f_1 as a distinct objective rather than collapsing to a two-objective (f_2, f_3) problem: it still discriminates the efficiency-extreme solutions (Table 4) and makes the efficiency–equity tension explicit—indeed it is the minimum-f_1 policy that dominates FIFO, which a two-objective reduction would obscure.

What governs performance. The ladder of Section 6.4 yields an empirical regularity we expect to recur: when a hard-constrained combinatorial problem is solved through a feasibility-preserving decoder, performance is dominated by the decoder and by matching the search operators to the *induced* representation—not by the metaheuristic family. A tuned real-coded NSGA-II falls *below* blind sampling because its interpolating operators are near-identity on the decoded permutation ($\tau = 0.99$; Section 6.4) and so barely search the combinatorial space; switching to permutation-native operators lifts three distinct paradigms (a GA,

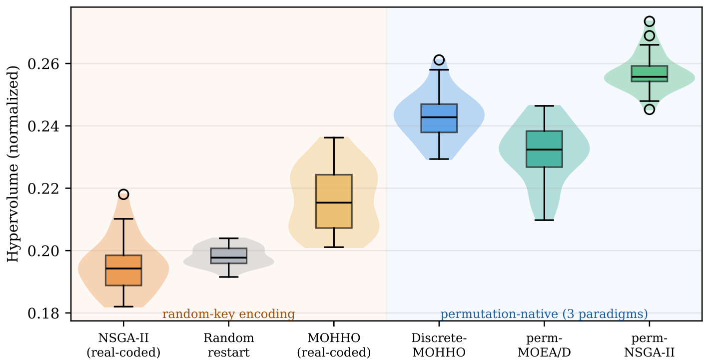


Fig. 9. The ladder on the multidimensional knapsack (MOMKP), one of the four structures of Table 8. As on the visa problem, all three random-key methods (NSGA-II, the MOHHO swarm, and random restart) sit below the three permutation-native methods—a clean two-tier split. The split reverses only on the *sequencing* landscapes (TSP, flow-shop), where the real-coded methods become competitive (Table 8).

a decomposition method, and a Harris Hawks optimizer) to the top, within about 1 % of one another. The practical guidance is twofold: invest design effort in the decoder and the representation before the search heuristic; and when a single dependable run matters—as in decision support—prefer the most stable optimizer, here Discrete-MOHHO.

Policy implication. The front is a quantified map of compromises for a decision-maker, from prioritizing efficiency ($f_1 = 8.79$) to radical equity ($f_2 = 2.0$) or full utilization ($f_3 = 0$). Notably, reducing disparity from 13.0 to 2.0 years costs only about 2.4 % in f_1 relative to the FIFO baseline, indicating that large equity gains are attainable at modest efficiency cost—consistent with proposals to reform the per-country caps [8,12]. Each front solution maps to a concrete allocation. Figure 10 shows the per-country reallocation of one balanced, full-utilization policy ($f_3 = 0$, $f_2 = 7.59$ years) against FIFO: it keeps the highest-demand countries (India, China, and the Philippines) at their 7 % ceiling and rebalances the rest—the residual “Rest of World” pool and several lower-served countries (Germany, Canada, Iran, and Nigeria) gain visas, drawn chiefly from the over-allocated Mexico and United Kingdom queues—an interpretable, auditable reallocation rather than an opaque score.

Limitations. Four caveats bound the conclusions. The model covers a single fiscal year and ignores intertemporal backlog dynamics; it uses 21 representative countries rather than the full roster; demand n_g is calibrated to published aggregate data [3,20,21] (although Section 6.5 shows the comparison is stable under ±20 % demand perturbations); and it assumes every assigned visa is used (no post-allocation refusals). These limitations affect external validity, not the

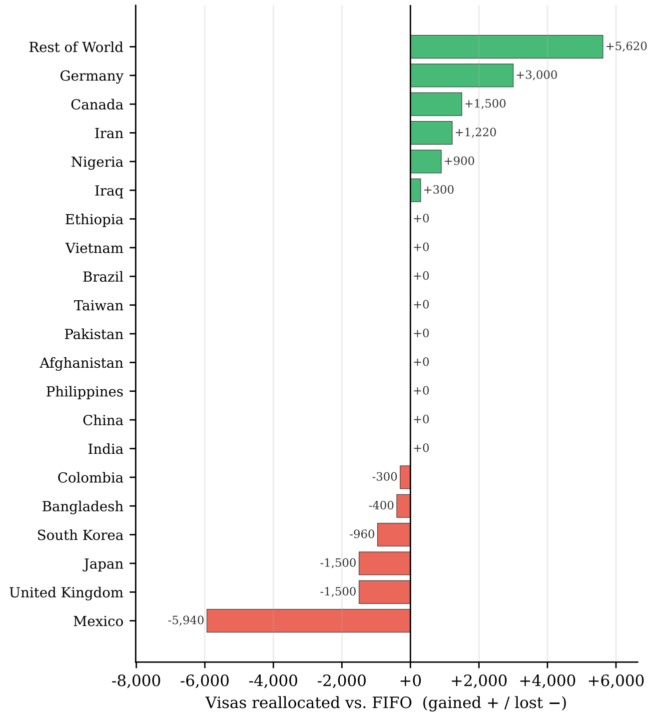


Fig. 10. Policy output: visas reallocated per country by a balanced full-utilization MOHHO solution relative to the FIFO baseline (gained in green, lost in red). The highest-demand countries remain at their per-country ceiling; the gains accrue to the residual pool and several lower-served countries, drawn chiefly from over-allocated queues (notably Mexico).

internal comparison among the six methods, which all run on identical data with an identical decoder and budget.

8 Conclusions

We formulated the annual allocation of U.S. employment-based visas as a three-objective integer program with six hard constraints and solved it through a feasibility-preserving SPV + greedy decoder that guarantees feasibility by construction—no penalties, no repair. A Taguchi L9 design replaced empirical parameter choices with a confirmed operating point. Across 30 independent runs the recovered Pareto front dominates the FIFO incumbent—chiefly by eliminating waste while matching its waiting load and maximal disparity—with 82 zero-waste solutions. Our central result comes from a controlled six-method ladder: the decoder alone (random restart) already beats *both* real-coded searches (NSGA-II and the MOHHO swarm), and—the lesson—the representation–operator match, not the metaheuristic family, governs performance. Matching operators to the representation lifts three distinct paradigms (a GA, a decomposition method, and a Harris Hawks optimizer) to within about 1 % of one another and above every random-key method; the Discrete-MOHHO we introduce improves on classic MOHHO by 4.7 % ($p = 9.3 \times 10^{-10}$) and is the most stable optimizer of all (CV 0.54 %), within 0.5 % of the best. All conclusions hold across five perturbed-demand instances, and a four-structure study (knapsack, TSP, and flow-shop scheduling) sharpens what generalizes: a representation-matched operator is the best optimizer on *every* structure and the metaheuristic family is second-order among matched methods (the robust, transferable law), whereas the spectacular collapse of a tuned real-coded GA below blind sampling occurs only on *selection* landscapes (visa, knapsack)—on *sequencing* landscapes (TSP, flow-shop) the same near-identity operators act as useful local search and the GA is competitive. A three-axis robustness analysis—non-saturating decoders, reference-point sweeps, and an exact MILP front—shows the representation effect is not an artifact of the decoder's budget saturation but only amplified by it, modest on this f_2-dominated instance and carried, as a general law, by the four structures. Operator order-preservation (τ=0.99) is problem-independent; whether it freezes or refines is landscape-dependent, and the precise determinant (a single search-headroom scalar does not capture it; $\rho = -0.90$ at $n = 5$, inconclusive) we report as an open question. Beyond the numbers, the optimizer yields an auditable *menu* of feasible trade-off allocations: a decision-support artifact, not a policy prescription, given the single-period model and the representative (partly synthetic) instance. Future work includes a multi-period formulation, the full country roster, further problem domains beyond the four studied here, and additional optimizers such as SPEA2.

Ethical Considerations. Allocating immigrant visas across nationalities is normatively charged. Our equity objective f_2 encodes one specific, contestable notion of fairness (minimizing the inter-country gap in mean wait), and the formulation treats nationalities as allocation buckets—a modeling choice with real human stakes, not a neutral given. We therefore frame the recovered front as *decision support* that makes trade-offs explicit and auditable, never as an autonomous decision-maker: any real-world use would require legal review, deliberation over which fairness criterion to adopt, and consultation with affected stakeholders. The instance is partly synthetic (see Section 3), so the per-country figures illustrate method behavior and must not be read as recommendations about real applicants.

Data and Code Availability. The problem data, the optimizer, the Taguchi design, and the scripts that reproduce every figure and table will be released in a public repository upon acceptance (repository withheld for double-blind review). All randomized studies use fixed seeds—the diagnostic suite uses seed 1, and the 30-run comparisons use fixed seed blocks recorded in the code—so the script regenerates every reported number deterministically.

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